Lab Problem 2

Objective:

To implement polynomial regression and analyze the effect of polynomial degree (M) on model performance. You will fit models of varying complexity and observe underfitting, a good fit, and overfitting.

Dataset Used:

- **Name:** Synthetic Sine Curve
- **Generation:** You must generate your own training data.
    - **Step A:** Create 10 training points (N=10).
    - **Step B:** Generate x values uniformly spaced between 0 and 1.
    - **Step C:** Generate y values by first computing $y_{\text{true}} = \sin(2 \cdot \pi \cdot x)$ and then adding Gaussian noise with a standard deviation of 0.05.

Task:

You must perform the following steps in order:

Task 1: Generate Data

- Generate the 10 training data points $(x_{\text{train}}, y_{\text{train}})$
- Generate a separate, clean set of 100 x points, denoted as x_{test}, spaced evenly from 0 to 1. The corresponding true y values are computed without any noise according to: $y_{\text{test}} = \sin(2 \cdot \pi \cdot x_{\text{test}})$
- This dataset is used exclusively for plotting the true underlying curve.

Task 2: Model Implementation and Training

- You will fit four separate models. For each of the following polynomial degrees **M = [0, 1, 3, 9]**:
    - Use sklearn.preprocessing Polynomial Features to transform your x_{train} data into the polynomial feature space.

- Initialize a sklearn.linear_model.LinearRegression model.
- Train the linear regression model on your transformed polynomial features and the y_{train} data.

Task 3: Model Evaluation

- For each of your four trained models (M=0, 1, 3, 9):
  - Calculate the **Root Mean Squared Error (RMSE)** on the **training data**.
  - Generate predictions for the "clean" 100 x_{test} points (remembering to transform them using the same Polynomial Features object).

4. Deliverables

Submit the following results from your implementation:

1. **RMSE Table:**
   a. (Provide a table showing the training RMSE for each degree M).

Polynomial Degree (M)	Training RMSE
0	(Your Value)
1	(Your Value)
3	(Your Value)
9	(Your Value)

2. **Model Fit Plot:**
   a. (Provide a single plot that clearly displays the following six elements):
      i. The 10 training data points (as blue scatter points).
      ii. The true sine curve (plotted using $x_{\text{test}}$and y_{test}, as a black line).
      iii. The M=0 model's predictions (as a line).
      iv. The M=1 model's predictions (as a line).
      v. The M=3 model's predictions (as a line).
      vi. The M=9 model's predictions (as a line).

   b. Ensure your plot has a legend to identify each line.